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RESEARCH ARTICLE

Feasibility of Using Beam Steering to Mitigate Ku-Band LEO-to-GEO Interference

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ABSTRACT Low-Earth orbiting (LEO) satellites now hold promise to provide broadband service anywhere in the world, but they must share the radio spectrum with geosynchronous Earth-orbiting (GEO) satellites. The United Nations International Telecommunications Union (ITU) and the U.S. Federal Communications Commission (FCC) require that operators of LEO satellite systems avoid interfering with GEO satellite systems even though both use the same bands within the radio spectrum. Such interference can occur when a LEO satellite passes through an area between a GEO earth station and its intended GEO satellite. This is called an in-line event. In this paper we develop a first order estimate of the probability of occurrence of such an event, an indication that Ku-band LEO-to-GEO interference is a serious problem. We describe the use of beam steering to mitigate such interference and consider its feasibility for currently deployed LEO constellations. We find that beam steering appears to be a practical and effective approach for a LEO constellation that has this capability and uses sufficiently narrow beams.

INDEX TERMS Communication satellite, in-line event, interference, LEO, NGSO.

I. INTRODUCTION

Non-geosynchronous orbit (NGSO) satellites are coming into widespread use, orbiting the Earth below the 35,000 km altitude of geosynchronous satellite orbit (GSO) satellites. GSO satellites, also called geosynchronous Earth orbit (GEO) satellites, orbit the Earth once every 24 hours and, when seen from the Earth's surface, stay at a fixed position in the sky. NGSO satellites, including medium Earth orbit (MEO) satellites at altitudes between 2000 and 35,000 km and low Earth orbit (LEO) satellites at altitudes below 2000 km, orbit the Earth more frequently and, when seen from the Earth's surface, move across the sky.

GEO satellites provide important services, but they may work poorly for interactive applications. This is because of the high latencies resulting from the propagation delay of radio signals traveling between earth stations and the high-altitude satellites. Fig. 1 shows a number of GEO and LEO satellites. The lower altitude of the LEO satellites results in much lower latencies than those of the GEO satellites.

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New LEO satellite systems promise to provide broadband service with low latencies and high throughputs throughout the world, including regions where the density of likely subscribers is low and terrestrial infrastructure is unable to support broadband service at a reasonable cost per user.

Several companies now have broadband constellations deployed or planned. These include SpaceX, with its Starlink project, OneWeb, Telesat, and Amazon. Starlink is farthest along with more than 2000 of a planned 4236 satellites in orbit. OneWeb has 358 of a planned 648 satellites in orbit. Telesat and Amazon are also planning to deploy LEO constellations to provide worldwide broadband service. Telesat plans to launch 1671 satellites, and Amazon plans 3236 satellites [1], [2].

II. RADIO INTERFERENCE

A challenge for current and future constellations of LEO satellites is that they must share spectrum in the Ku-, Ka- and V-bands with GEO satellites and with each other. A GEO earth station that must receive signals from a GEO satellite 35,000 km overhead cannot tolerate a great deal of interference. Nevertheless, it is the responsibility of the

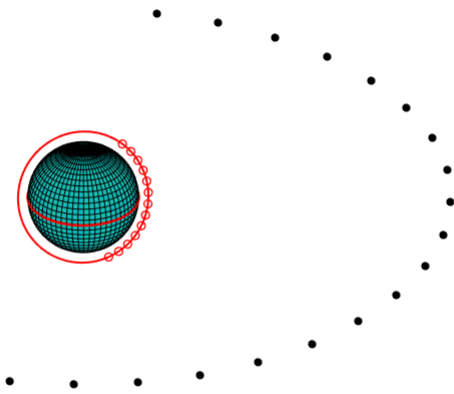


FIGURE 1. LEO satellites (shown in red) orbit at a much lower altitude than GEO satellites (shown in black) at 35,000 km.

operators of LEO satellites to protect GEO systems from harmful interference. The International Telecommunications Union (ITU), which is the telecommunications arm of the United Nations, has addressed this interference risk in its regulations and requires that LEO satellite operators avoid interfering with GEO systems. Under ITU regulations [3], “non-geostationary-satellite systems shall not cause unacceptable interference to ... geostationary satellite networks in the fixed-satellite service and the broadcasting-satellite service.” Unacceptable interference is defined in [3] in terms of effective power flux density (EPFD) levels that must not be exceeded at a GSO station. The operator of an NGSO satellite is free to determine the means by which the satellite will avoid causing harmful interference, which could include adjusting transmit power, adjusting beam direction, or temporarily stopping transmission. These same ITU regulations also state that the operator of an NGSO satellite “shall not claim protection from geostationary satellite networks,” which means that GSO satellites are not expected to alter their design or operation in any way to accommodate NGSO satellites. This is typical of *primary-secondary spectrum sharing* [4], where in this case the GSO satellites are primary.

National spectrum regulators play an important role in enforcement. Many national regulators, such as the Federal Communications Commission (FCC) in the United States, have adopted rules that simply incorporate the ITU regulations as described in [5]. This means that a LEO satellite that was licensed by the FCC and that violates these ITU regulations by causing harmful interference to a GEO satellite has also violated U.S. regulations and that the FCC is empowered to enforce those regulations with fines. If a LEO operator wants to serve users in nations with big commercial markets, that operator will need to work with national regulators that adopt the ITU policy.

Thus, LEO systems must coexist and share Ku-, Ka-, and V-band spectrum with each other, and they must not cause objectionable interference to GEO systems. Preventing LEO-to-GEO and LEO-to-LEO interference poses serious technical challenges [6], [7]. One recently described

approach to both problems called “look aside,” imposes a minimum separation angle between a “victim” (protected) satellite system and all potentially interfering links from other systems [7]. This approach is intended to protect all GSO systems from interference by NGSO systems and to protect each NGSO system from interference from other NGSO systems.

Absent further guidance from the ITU and national regulators like the FCC, there will likely be a need to implement “look aside” or a similar approach. But such approaches can entail operational complexity. Determination of the minimum separation angle needed to prevent objectionable interference will have to consider transmit powers and antenna gains of existing and proposed GSO and NGSO systems [6], [7]. There has been some early work on NGSO-to-GSO interference [8], [9], but as pointed out in [7], this work does not consider dense NGSO deployments like the ones that are now in operation and planned.

Because two LEO constellations (Starlink and OneWeb) are now partially deployed, there is some urgency to these issues. LEO operators may choose to separate their own systems by frequency and/or polarization, but they will still be sharing spectrum with the GEO systems that they are required to protect. The problem is particularly urgent at Ku-band because currently deployed OneWeb and Starlink systems use Ku-band to link to their user terminals [2], and Ku-band also is widely used by GEO systems, including service to many small earth stations used for Internet access and to receive television broadcasts.

A LEO satellite system can interfere with a GEO satellite system during an in-line event, an event where a LEO satellite, as it traverses its orbit, passes through an area between a GEO earth station and its intended GEO satellite. Interference can occur when the LEO satellite is either on a line between the GEO earth station and GEO satellite or when it is nearly in line and in an area determined by the GEO earth station’s beamwidth.

Here we focus on the risk of LEO-to-GEO interference and its mitigation at Ku-band. We first estimate the probability that a single GEO earth station will experience an in-line event as an indication of the seriousness of the problem with currently deployed LEO constellations. Then we describe a simple beam steering approach to prevent LEO-to-GEO interference. The approach protects all GEO systems without the need for a database or other information about the details of the GEO systems, e.g., satellite orbital positions, earth station locations, frequencies, transmit powers and antenna polarizations.

III. ESTIMATING THE RISK OF LEO-TO-GEO INTERFERENCE

We consider the seriousness of the LEO-to-GEO interference problem by estimating the probability that at any instant an individual GEO earth station will experience an in-line event caused by an orbiting LEO satellite. If both satellites transmit at the same time in the direction of the line that passes through both satellites and the GEO earth station, and if they use an

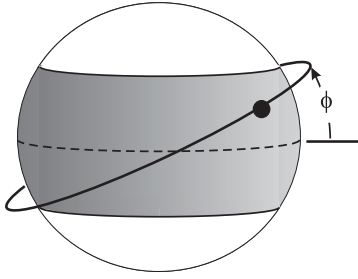


FIGURE 2. LEO satellite's orbital inclination and coverage.

overlapping set of frequencies, then interference can occur. Depending on the magnitude of parameters such as transmit power, path loss, and antenna gain for the two systems, this interference can be harmful. Thus, the probability of occurrence of an in-line event at any instant is an important first-order indicator of the seriousness of the LEO-to-GEO interference problem.

Although all GEO satellites orbit along an arc that circles the earth above the equator, LEO satellites often use inclined orbits. As shown in Fig. 2, the orbit's angle of inclination Φ is the angle between the LEO's orbital plane and the Earth's equatorial plane. An orbit whose orbital inclination $\Phi = 90^\circ$ is called a polar orbit. Satellites in polar orbit can provide coverage to all points on the surface of the Earth, but this is not true of orbits with other inclinations. Satellites in inclined orbits like the one shown in Fig. 2 cover only locations shown in the shaded area between northern and southern coverage limits.

We can model the Earth as a sphere with radius R_E and surface area,

$$A_E = 4\pi R_E^2. \quad (1)$$

Similarly, we can model the surface area of a polar orbiting LEO constellation's orbital shell as a sphere with radius R_O and surface area A_O ,

$$A_O|_{max} = 4\pi R_O^2, \quad (2)$$

where the *max* subscript indicates that this is the maximum value A_O can attain.

If however, the LEO constellation has an inclined orbit with inclination Φ , the LEO satellite shell does not reach the caps of the spherical shell, the parts of the orbital shell outside the shaded area of Fig. 2. The area of each cap A_C is,

$$A_C = 2\pi R_O^2(1 - \sin \Phi). \quad (3)$$

And the surface area of the inclined orbital shell A_O is therefore reduced by twice the area of each polar cap. From (2) and (3),

$$\begin{aligned} A_O &= A_O|_{max} - 2A_C \\ &= 4\pi R_O^2 - 2[2\pi R_O^2(1 - \sin \Phi)] \\ &= 4\pi R_O^2 \sin \Phi. \end{aligned} \quad (4)$$

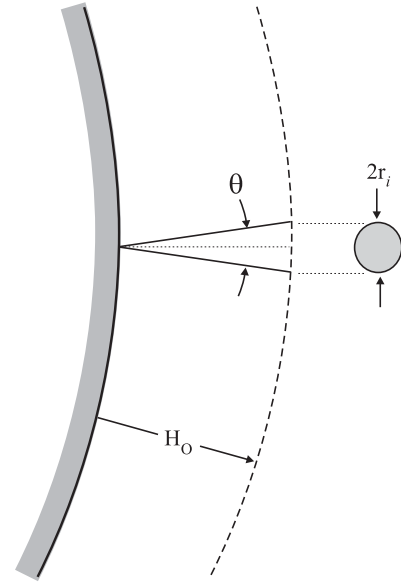


FIGURE 3. Interference zone with a GEO earth station at the equator and at the same longitude as its intended GEO satellite.

As discussed in the introduction, under FCC and ITU regulations, operators of LEO constellations are required to avoid interference with GEO satellite systems. Serious LEO-to-GEO interference can occur when the transmissions of a LEO satellite interfere with transmissions from a GEO satellite to a GEO earth station (the GEO downlink). Interference can occur during an in-line event, when the LEO satellite is either on a line between the GEO earth station and GEO satellite or when it is nearly in line and in an area defined by the beamwidth of the GEO earth station antenna. Here we call this the *interference zone*, the area in which there is the potential for the LEO satellite to cause objectionable interference.

First, we consider a GEO earth station located at the equator and aimed at a GEO satellite that is directly overhead. To calculate the interference zone's area A_I , we model the beam of the earth station as a cone, where the angle between the cones' edges θ represents the beamwidth of the antenna.

A constellation of LEO satellites orbit at nearly equal altitudes and occupy a thin shell, whose face is a sphere. Fig. 3 portrays the situation using a planar cross section that contains the GEO earth station, its intended GEO satellite, and the Earth's north and south poles. The GEO earth station antenna's beam is pointed vertically toward its intended satellite, but before it reaches the GEO satellite, the conical beam of the GEO earth station intersects the LEO orbital sphere and creates an interference zone.

In Fig. 3 we approximate a small portion of the orbital sphere as a plane. The interference zone at the intersection of the cone and this plane is a conic section, and since the axis of the cone is perpendicular to the plane, this interference zone is a circle. The radius r_i of the circle is,

$$r_i = H_O \tan \left(\frac{\theta}{2} \right), \quad (5)$$

where H_O is the LEO's altitude and θ is the beamwidth of the GEO earth station antenna. The interference zone's area A_I is the area of this circle,

$$\begin{aligned} A_I &= \pi r_i^2 \\ &= \pi H_O^2 \tan^2 \left(\frac{\theta}{2} \right). \end{aligned} \quad (6)$$

If the LEO satellites are in polar orbit and uniformly distributed within their orbital shell, we can estimate the probability P_I that a single LEO satellite will be in the GEO earth station's interference zone (an in-line event) as the interference zone's area divided by the area of the LEO spherical shell,

$$\begin{aligned} P_I &= \frac{A_I}{A_O} \\ &= \pi H_O^2 \tan^2 \left(\frac{\theta}{2} \right) / 4\pi R_O^2 \\ &= \frac{H_O^2}{4R_O^2} \tan^2 \left(\frac{\theta}{2} \right). \end{aligned} \quad (7)$$

If the LEO satellites are uniformly distributed in any orbit (inclined or polar) with inclination Φ , the area of the LEO orbital shell will be as given in (4). Thus, for any LEO orbit the probability P_I that a single LEO satellite will be in the GEO earth station's interference zone can likewise be estimated as,

$$\begin{aligned} P_I &= \frac{A_I}{A_O} \\ &= \pi H_O^2 \tan^2 \left(\frac{\theta}{2} \right) / 4\pi R_O^2 \sin \Phi \\ &= \frac{H_O^2}{4R_O^2 \sin \Phi} \tan^2 \left(\frac{\theta}{2} \right). \end{aligned} \quad (8)$$

We now consider a LEO constellation in any orbit (inclined or polar) and with a GEO earth station located anywhere on the Earth's surface where it can "see" its intended GEO satellite. To do so, the GEO earth station will use a look angle λ , which is the elevation angle (above the horizon) at which it must direct its antenna.

To calculate the area of the interference zone in this case, we again model the beam of the earth station antenna as a cone, where the angle between the cone's axis and its edge represents half the beamwidth of the antenna.

Fig. 4 portrays the situation, again using a planar cross section that contains the GEO earth station and its intended GEO satellite. We model the portion of the Earth where the GEO earth station is located as a plane and a small portion of the orbital sphere as a parallel plane. The conical beam of the GEO earth station intersects the LEO orbital sphere and creates a conic section. Because the axis of the cone intersects the plane representing a portion of the orbital sphere at an oblique angle, the interference zone is an ellipse. While it is true that the plane representing a portion of the Earth's surface at the GEO earth station's location and the plane representing a portion of the LEO orbital sphere are not perfectly parallel,

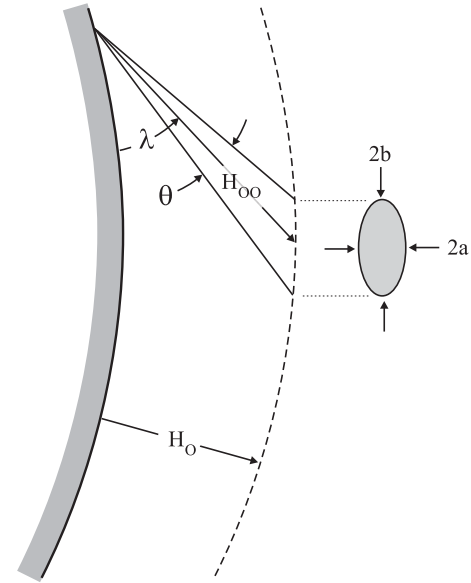


FIGURE 4. Interference zone with a GEO earth station at any latitude and longitude.

for high look angles, the error caused is small because the LEO altitude is small relative to the earth's radius. This simplification greatly reduces computational requirements while introducing only marginal errors to be later noted.

The distance from the GEO earth station to the LEO satellite H_{OO} is greater than H_O . It increases as the GEO earth station look angle λ decreases. But for high look angles, the distance H_{OO} can be approximated as,

$$H_{OO} \approx \frac{H_O}{\sin \lambda}. \quad (9)$$

The shape of the interference ellipse also varies with λ , becoming more eccentric as the look angle λ decreases. The length $2a$ of the ellipse's minor axis differs from the earlier circle's diameter because H_{OO} is greater than H_O .

Likewise, we can approximate a as,

$$a \approx H_{OO} \tan \left(\frac{\theta}{2} \right). \quad (10)$$

The ellipse's major axis, with length $2b$, is longer than that of the minor axis, with length $2a$, because the cone intersects the shell at an oblique angle, and the eccentricity of the ellipse increases with decreasing λ . The eccentricity e is given by,

$$e = \frac{\sin \beta}{\sin \alpha}, \quad (11)$$

where β is the angle at which the LEO orbital plane intersects a plane perpendicular to the cone's axis, and α is the angle between the edge of the cone and the plane perpendicular to the cone's axis.

The ratio of the ellipse's minor axis length to its major axis length is,

$$\frac{2a}{2b} = \frac{a}{b} = \sqrt{1 - e^2}. \quad (12)$$

Thus,

$$b = \frac{a}{\sqrt{1-e^2}}. \quad (13)$$

In this case,

$$\alpha = 90^\circ - \frac{\theta}{2}, \quad (14)$$

and

$$\beta = 90^\circ - \lambda, \quad (15)$$

so that,

$$\begin{aligned} e &= \frac{\sin(90^\circ - \lambda)}{\sin(90^\circ - \theta/2)} \\ &= \frac{\cos \lambda}{\cos(\theta/2)}. \end{aligned} \quad (16)$$

Half of the axes' lengths a and b can be calculated as,

$$a = H_{OO} \tan\left(\frac{\theta}{2}\right), \quad (17)$$

and

$$b = \frac{a}{\sqrt{1-e^2}}. \quad (18)$$

The area of an ellipse is πab , Thus, the area of the elliptical interference zone A_I is,

$$\begin{aligned} A_I &= \pi H_{OO}^2 \frac{\tan^2(\theta/2)}{\sqrt{1-e^2}} \\ &= \frac{\pi H_{OO}^2 \tan^2(\theta/2)}{\sqrt{1 - \frac{\cos^2 \lambda}{\cos^2(\theta/2)}}}. \end{aligned} \quad (19)$$

Since $H_{OO} = H_O / \sin \lambda$,

$$A_I = \frac{\pi H_O^2 \tan^2(\theta/2)}{\sin^2 \lambda \sqrt{1 - \frac{\cos^2 \lambda}{\cos^2(\theta/2)}}}. \quad (20)$$

This computation of A_I , based on the assumption that the plane representing a portion of the Earth at the GEO earth station's location and the plane representing the LEO orbital sphere at the LEO satellite's location are parallel, is a good approximation for realistic values of H_O and values of $\lambda > 45^\circ$. Indeed, for $H_O = 1000$ km, the error introduced by the planar model of Fig. 4 is on the order of only 25% for $\lambda = 45^\circ$, which is the maximum error for λ between 45° and 90° . For λ at a more typical angle of 60° , the resulting error is less than 10%. For $H_O = 500$ km, these errors are substantially smaller. Errors in this range are consistent with a first-order estimate of the probability of an in-line event.

We can now calculate the probability P_I ,

$$\begin{aligned} P_I &= \frac{A_I}{A_O} \\ &= \frac{\pi H_O^2 \tan^2(\theta/2)}{4\pi R_O^2 \sin \Phi \sin^2 \lambda \sqrt{1 - \frac{\cos^2 \lambda}{\cos^2(\theta/2)}}}, \end{aligned}$$

$$= \frac{H_O^2 \tan^2(\theta/2)}{4R_O^2 \sin \Phi \sin^2 \lambda \sqrt{1 - \frac{\cos^2 \lambda}{\cos^2(\theta/2)}}}, \quad (21)$$

with A_O and A_I as given in (4) and (20), respectively.

To summarize, P_I is the probability that a single LEO satellite is at any instant in the interference zone of a single GEO earth station, i.e., in a position to cause objectionable interference to that earth station.

The probability that a single GEO earth station will experience an in-line event resulting from an entire constellation of LEO satellites is directly proportional to the number of satellites in the constellation N if the LEO satellites are uniformly distributed in space. Thus P_{IN} , the probability of an in-line event, is equal to N times the probability of an in-line event resulting from a single LEO satellite P_I ,

$$P_{IN} = NP_I. \quad (22)$$

For realistic cases, the values of N , Φ , and λ do not result in cases where multiple LEO satellites simultaneously enter the interference zone, and (21) and (22) will give a good first-order estimate of the probability that a GEO earth station will experience an in-line event. Thus, if λ is restricted to values of at least 45° (which will be the case when GEO ground stations are located within the more inhabited parts of the world) and LEO satellites are distributed uniformly within the orbital shell, our results provide good estimates.

The most immediate concern regarding LEO-to-GEO interference is at-Ku-band (14/12 GHz). For this band the beamwidth (between first nulls) of a 0.4-meter antenna is about 10° . Such small GEO earth stations are widely used for broadcast television reception and Internet access. Although there are a variety of different such antennas in use, we use this beamwidth for our probability estimates. Larger earth stations will, of course, have narrower beamwidths. For example, the beamwidth of a 10-meter antenna is less than 1° . We focus here on small GEO earth stations because they represent a worst case, because there are so many of them in use, and because it is unlikely that all their locations are available in any database.

IV. INTERFERENCE RISK RESULTS

For purposes of illustration, we first consider a representative example in which a small Ku-band GEO earth station is located at mid-latitude and uses a look angle λ of 45° . In our example, one LEO constellation has been deployed with 1000 satellites in polar orbit at 1000 km altitude. We calculate that P_{IN} in this case is 10%, a remarkably high number because it means that just a single small GEO earth station can be expected to experience an in-line event 10% of the time, or 2.4 hours/day.

We note, however, that our results are dependent on the values we have used for θ , N , λ , H_O , and Φ . For other values, the probability of an in-line event will be:

- Smaller for a midsize or large earth station, either of which will have a smaller beamwidth θ ,

TABLE 1. In-line event probability first-order estimates for the OneWeb constellation.

LEO Orbital Altitude (km)	LEO Orbital Inclination (degrees)	Number of LEO Satellites in constellation	Probability of In-Line Event
1200	90	358	4.89%
1200	90	290	3.96%
TOTAL			8.84%

- Larger for a higher number of satellites N because there will be more satellites that can enter the interference zone,
- Larger for a GEO earth station at high latitude, where it will need to use a lower look angle λ , creating a larger interference zone,
- Smaller for a LEO constellation at an altitude H_O lower than 1000 km because such an altitude will result in a smaller interference zone, and
- Somewhat larger for LEO constellations with orbital inclinations Φ less than 90° because the orbital shell will be smaller, and satellites will be spaced more closely.

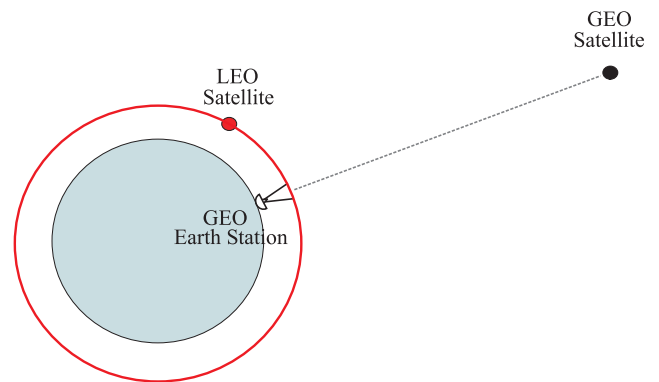
Nevertheless, we believe that the example with $P_{IN} = 10\%$ is a useful starting point. But we now move beyond the illustrative example and present results for the current and planned OneWeb and Starlink constellations. We estimate the probability of an in-line event for a small Ku-band GEO earth station at low- or mid-latitude with each of these constellations deployed.

Table 1 shows results for the OneWeb constellation, which currently has 358 satellites in polar orbit at an altitude H_O of 1200 km [10]. For the current constellation, we estimate the probability of an in-line event at 4.89%. OneWeb plans to launch 290 more satellites at 1200 km, with an in-line event probability of 3.96%. The final constellation will have 648 satellites, all in polar orbit at 1200 km [10]. For OneWeb's full constellation, we estimate the probability of an in-line event at about 9%. We give this result to its nearest integer value because it is a first-order estimate.

Table 2 shows results for the current and planned Starlink constellations. Starlink now has more than 2000 satellites in orbit [11]. They plan four orbital shells, as shown in Table 2. The first shell, which is now complete, comprises 1584 satellites orbiting at an altitude of 550 km at an orbital inclination of 53° . Three more shells are being added, the next with 1584 satellites orbiting at 540 km at an orbital inclination of 53.2° , another with 720 satellites orbiting at 570 km with an orbital inclination of 70° , and the final shell with 348 satellites orbiting at 560 km in a near polar orbit with inclination 82.4° [12]. As shown in Table 2, we estimate the probabilities of an in-line event for each of these four shells at approximately 6.81%, 6.56%, 2.81%, and 1.25%. The total for all four shells is about 17%. We again give this result to its nearest integer value because it is a first-order estimate.

TABLE 2. In-line event probability first-order estimates for the Starlink constellation.

LEO Orbital Altitude (km)	LEO Orbital Inclination (degrees)	Number of LEO Satellites in constellation	Probability of In-Line Event
550	53	1584	6.81%
540	53.2	1584	6.56%
570	70	720	2.81%
560	82.4	348	1.25%
TOTAL			17.42%

**FIGURE 5.** LEO satellite will sometimes be in-line with a GEO earth station and its intended GEO satellite.

V. MITIGATING KU-BAND LEO-TO-GEO INTERFERENCE

How can LEO-to-GEO satellite interference be mitigated? As shown in Fig. 5, when an orbiting LEO satellite is in-line with a GEO earth station and its intended GEO satellite, interference can occur. The LEO satellite might avoid such interference by ceasing transmission, or if it knows the frequencies being used on the GEO link, it might switch to a different frequency. To know that it is in-line, a LEO satellite would need to have information on the locations of all vulnerable GEO earth stations and the orbital positions of their intended GEO satellites. But it is unlikely that such information is readily available, particularly information on the many small GEO earth stations used for television broadcast reception and Internet access.

We describe the use of straightforward beam steering with electronically steerable phased array antennas to avoid interference. Stated simply, LEO satellites and earth stations avoid steering their beams in directions that could cause interference to all GEO earth stations and satellites. With this beam steering approach, LEO operators do not need to know the orbital positions and frequencies of individual GEO satellites or the locations and frequencies of individual GEO earth stations.

We describe both uplink and downlink beam steering. A LEO earth station will use uplink beam steering as it transmits to establish and maintain a connection with a LEO satellite. A LEO satellite will use downlink beam steering for

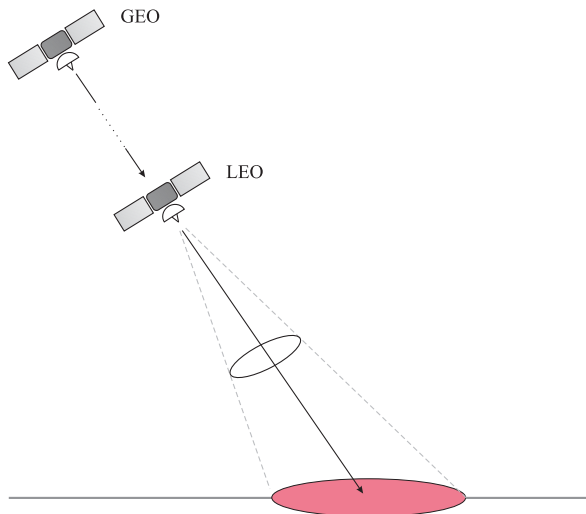


FIGURE 6. LEO satellite's forbidden zone is formed by the intersection of a cone representing its beamwidth and the Earth's surface.

the same reason and as it transmits beacon signals to make LEO earth stations aware of its presence and availability.

A LEO earth station can avoid causing uplink interference by not pointing its beam toward or near the GEO arc, thus avoiding interference to any GEO satellite. Based on the LEO earth station's antenna beamwidth, one can establish its *forbidden zone*, i.e., the directions in which the LEO earth station must not point a beam. This zone can also be called an "exclusion zone". To be conservative, we suggest using the antenna's first null beamwidth and adding a margin of safety to ensure the elimination of any interference to vulnerable GEO satellites. This will limit the number of LEO satellites available to the LEO earth station at any moment, but the earth station will have other satellites in its field of view and the ability to communicate with any of them.

On the other hand, a LEO satellite can avoid downlink interference to GEO earth stations by not pointing a beam toward any GEO earth station in danger of experiencing an in-line event. This is illustrated in Fig. 6, where we draw a line from a GEO satellite to a LEO satellite at a position along its orbit. We then extend this line to the surface of the Earth but surround it by a cone that accounts for the antenna beamwidth of the LEO satellite. Again, to be conservative, we suggest using the antenna's first null beamwidth and adding a margin of safety to ensure the elimination of any interference to vulnerable GEO satellites. The cone intersects the surface of the Earth in a near ellipse that is this LEO satellite's *forbidden zone* (shown in red) that will protect the GEO satellite when the LEO satellite is in this position. If the LEO satellite avoids pointing its beam into this forbidden zone, it will not cause interference to any GEO earth station on the ground using this GEO satellite.

This is further illustrated in Fig. 7, which shows a LEO satellite at a position in an orbit with 53° inclination protecting a GEO satellite at a position along the GEO arc and west

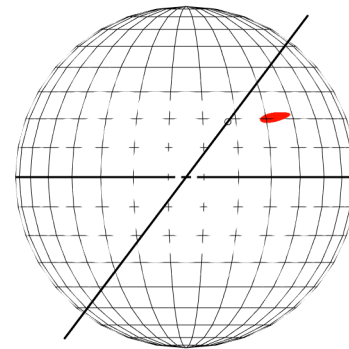


FIGURE 7. LEO satellite's forbidden zone needed to protect a single GEO satellite to its west.

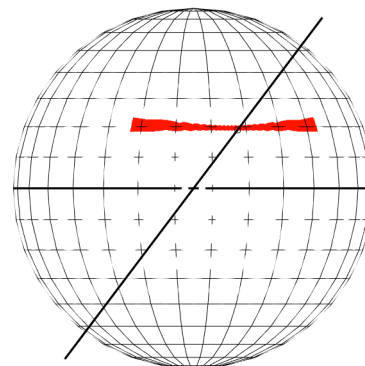


FIGURE 8. LEO satellite's forbidden zone needed to protect all GEO satellites.

of the LEO satellite. We pass a line from the GEO satellite to the LEO satellite and extend this line to the surface of the Earth surrounded by the cone we have described. The cone intersects the surface of the Earth in a near elliptical area, which is the forbidden zone (shown in red) that will protect this GEO satellite when the LEO satellite is in this position. (In this example and those that follow, we assume a LEO satellite with a beamwidth including a safety margin totaling 10° and in an orbit at 1200 km altitude.)

Because we need to protect all GEO satellites, we repeat this process for every GEO satellite position along the GEO arc, giving the complete forbidden zone for the LEO satellite in this position. (Fig. 8) We carry out the same process for a LEO satellite at every point in its orbit to find forbidden zones for all LEO satellite positions. Fig. 9 shows forbidden zones for four positions of the LEO satellite in the northern hemisphere. When the LEO satellite is over the equator, the forbidden zone comprises a band centered on the equator. But as the LEO satellite moves north, the forbidden zone also moves north and gradually moves from beneath the LEO satellite to positions farther and farther north of the satellite. As seen in Fig. 10, the situation is similar for LEO satellite positions in the southern hemisphere. In this case, as the LEO satellite moves farther south from the equator, the forbidden zone also moves south but gradually moves from beneath

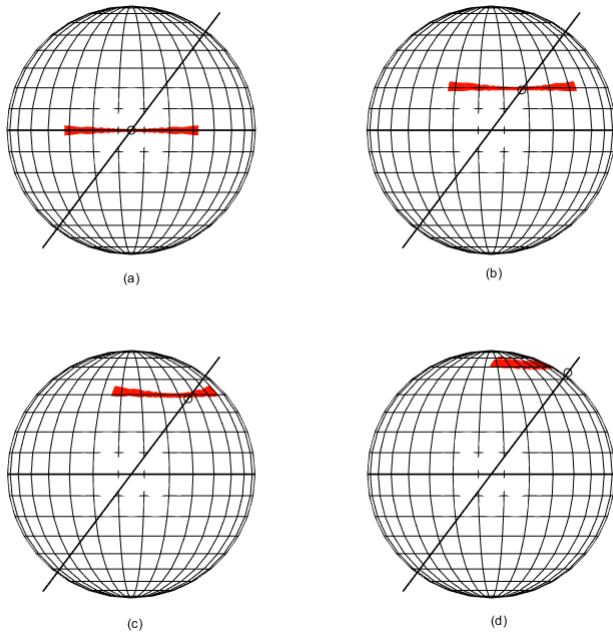


FIGURE 9. Satellite's forbidden zone, shown in (a)-(d), moves as the satellite traverses its orbit in the northern hemisphere.

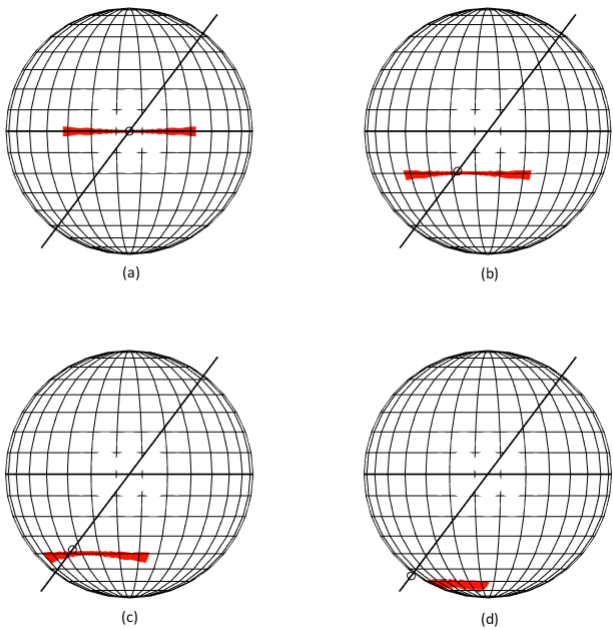


FIGURE 10. LEO satellite's forbidden zone, shown in (a)-(d), moves as the satellite traverses its orbit in the southern hemisphere.

the LEO satellite to positions farther and farther south of the satellite.

Fig 11 shows forbidden zones for a polar orbit. The result is similar to the result for a LEO with 53° orbital inclination. When the LEO satellite is over the equator, the forbidden zone comprises a band centered on the equator, and as the LEO satellite moves north, the forbidden zone also moves north but

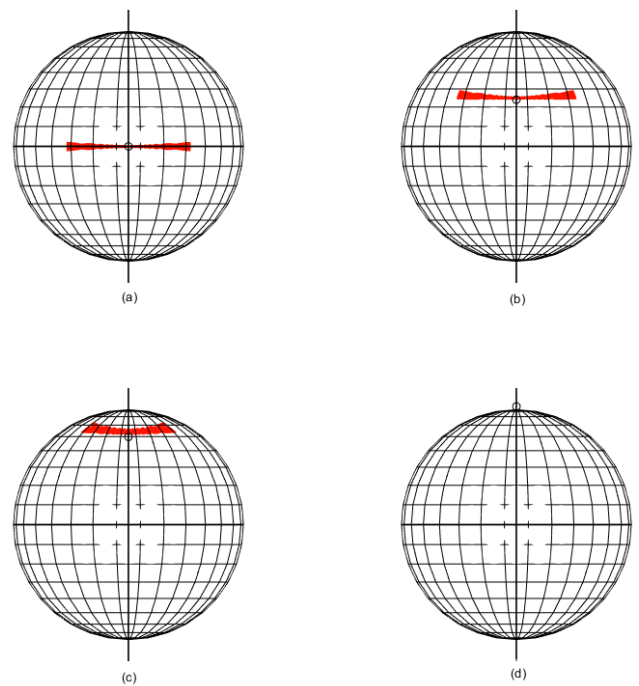


FIGURE 11. LEO satellite in polar orbit has a forbidden zone, shown in (a)-(d), that moves as the satellite traverses its orbit in the northern hemisphere.

gradually moves from beneath the satellite to positions farther and farther north of the satellite. As the satellite approaches the North Pole, the cone descending from the LEO satellite completely misses the Earth's surface, and there is no forbidden zone at all.

VI. CONCLUSION AND DISCUSSION

Our first order in-line event probability estimates indicate that the risk of LEO-to-GEO satellite interference is high. With the currently deployed and planned OneWeb and Starlink constellations, the probability at any moment that a single small GEO earth station is in danger of experiencing a OneWeb-caused outage is about 9%. The probability that it is in danger of experiencing a Starlink-caused outage is about 17%. These probabilities are for just a single small GEO earth station. They are consistent with the conclusion of [7] that LEO-to-GEO interference is a serious problem, and it is clear that mitigation measures are needed.

Simple beam steering can be a useful way for LEO constellations to share spectrum with GEO satellite systems. Interference mitigation on the uplink can be accomplished by requiring LEO earth stations to steer their beams away from the GEO orbital arc. To mitigate downlink interference, we suggest that LEO satellites, even when transmitting beacon signals, avoid steering their beams into the forbidden zones that we have described and illustrated with examples.

In our Ku-band beam steering examples (shown in Figs. 7-11), we have used conservative assumptions about

LEO satellites' altitudes and antenna beamwidths, including a satellite altitude of 1200 km, which is the highest orbital altitude of any of the shells in the two deployed LEO constellations [1], [2]. An orbital altitude of 1200 km represents the worst case because LEO satellites in lower orbits will have smaller forbidden zones. Our examples use a LEO satellite antenna beamwidth (including safety margin) of 10° . Starlink has electronic beam steering capability with beamwidths well under 10° and orbital altitudes far less than 1200 km [1], [2]. Thus, simple beam steering is a feasible approach for Starlink.

But OneWeb satellites use fixed antennas that have larger, highly elliptical beams and do not have beam steering capability [1], [2], [13], [14]. The company has said that it will use "progressive pitch" to avoid GEO interference. Although the details have not been made available, one possible progressive pitch approach that uses a combination of tilting satellites and turning off beams has been described in [14]. But with its current constellation, OneWeb is unable to use the simple beam steering approach we have described.

The aggregate effective power flux density (EPFD) received by a GEO earth station from all LEO satellites in a constellation must not exceed the limit set by the ITU and FCC [3], [5], [6]. Where beam steering is used to mitigate interference, the forbidden zones must be large enough to assure that this aggregate EPFD limit is not exceeded [6]. But inevitably a LEO satellite will radiate some energy outside its main beam, creating the danger that, for a large constellation, the aggregate EPFD resulting from such radiation will exceed the limit. As the size of a LEO constellation increases, it will be necessary to increase the size of its forbidden zones in order to limit the aggregate EPFD received by a GEO earth station. With the details of each LEO constellation's terminal and satellite transmit powers and antenna patterns, as well as its orbital characteristics, it will be possible to determine the sizes of the needed forbidden zones for a given constellation size, but to our knowledge, antenna patterns are not yet publicly available for the Starlink constellation.

Although the sizes of the forbidden zones can be increased as more satellites are added to a constellation, this may reduce system performance as fewer satellites are available to each LEO earth station and fewer earth stations are available to each LEO satellite. At some constellation size it may no longer be possible to continue to increase forbidden zone sizes and still maintain acceptable system performance. For example, Starlink plans to add many more satellites to its constellation beyond the 4236 satellites in its current launch campaign. At some constellation size, simple beam steering will no longer be an effective way to limit the aggregate EPFD of the constellation while maintaining acceptable system performance. With detailed information on Starlink's constellation, or on any other LEO constellation, it will be possible to explore the trade off between constellation size and system performance.

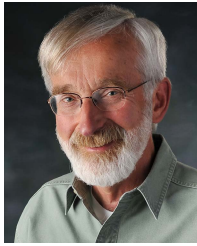
But even if each LEO constellation properly limits its own aggregate EPFD, with no coordination between two constellations, e.g., Starlink and OneWeb, their combined aggregate EPFD may still exceed the ITU and FCC limit. But if band-splitting is used to give each LEO system its own spectrum assignment, the aggregate EPFD limit will apply to each LEO constellation separately, and each LEO operator can design its own beam steering system without regard to interference caused by other LEO systems.

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